

AgentFlow: An Investigative Visual Analytics Platform for Reconstructing the TenantThread Embargo Breach (VAST Challenge 2026, MC-1)

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ABSTRACT

Mini-Challenge 1 of the VAST Challenge 2026 asks how embargoed information about the *Project HarborCrest* merger (TenantThread–CivicLoom) reached the public FleX platform ~35 minutes before the 18:00, June 5, 2046 embargo lift, given the communication log of seven autonomous agents. We present **AgentFlow**, a forensic investigation console — deliberately not a dashboard — built with D3.js v7 on a reproducible Python feature-engineering pipeline. The investigation is condensed into two linked views: the *Behavioral Drift Braid*, which plots each agent as a thread on an ordinal covert-to-public posture ladder composed with expected baselines, covert-coordination arcs, compliance gates, the causal chain E01–E12, and a leading-indicator seismograph with pre-crisis $\mu+1\sigma/\mu+2\sigma$ thresholds; and the *Hypothesis Balance*, a transparent evidence model whose twelve categories, weighted with diminishing returns $w_c = b_c(1 - e^{-n_c/s_c})$, physically tilt a scale between INTENTIONAL and SYSTEMIC FAILURE. Every signal is a documented arithmetic formula, auditable down to the original message from a persistent Evidence Inspector. Over the 912 messages the balance tilts 25.3 vs. 14.2 toward a breach with a deliberate core, executed by the two agents who dominated the covert channel and preceded by early indicators detectable from May 22.

Index Terms: Visual analytics, behavioral anomaly detection, communication forensics, evidence weighting, VAST Challenge.

1 INTRODUCTION

TenantThread secretly negotiates its acquisition by CivicLoom under an embargo until 18:00, June 5, 2046. Seven autonomous agents (legal, social media, quality, PR, two interns, and a compliance agent, *The Judge*) exchange 912 messages over six channels of decreasing traceability; the embargoed fact surfaces publicly 35 minutes early. The analytic difficulty is threefold: evidence is scattered across messages, private reasoning (*internal_state*) and 77 public posts, sampled irregularly (daily, then hourly on the crisis day); no single message constitutes the breach — the signal is a distributed gradient of covert coordination, boundary testing and incremental disclosure via “defensible” governance language; and the central question (deliberate leak vs. systemic failure) is a judgment call. Our contributions: (1) a fully documented pipeline in which every score is an auditable formula; (2) the *Behavioral Drift Braid*, turning behavioral drift into literal geometry; (3) a composite *seismograph* with statistically grounded early-warning thresholds; (4) the *Hypothesis Balance*, an explicit evidence-weighting model that supports — not replaces — the analyst’s judgment.

2 DATA AND FEATURE ENGINEERING

One script derives seven JSON datasets (messages, rounds, network, actors, posts, events, meta). *No score is a black box*. Five transparent lexicon families flag both public content (nlp: what

was *said*) and private reasoning (nlp_inner: what was *known*): **merger** (the embargoed subject: civicloom, harborcrest, definitive agreement), **embargo** (awareness: 6 pm, confidential, material nonpublic), **execution** (the act: executing, going live), **compliance** (rationalization: 10b-5, defensible), and **governance** — the case-specific insight: the breach leaked through individually defensible “governance clarifications” (audit, consent management, access control), not blunt merger statements. Message sensitivity is

$$S(m) = 3.0|M_m| + 2.0|E_m| + 1.5|G_m| + 1.0|X_m| \quad (1)$$

over the merger/embargo/governance/execution term counts, weights decreasing with legal gravity of each family. An *embargo violation* is the strict conjunction $V(m) = (\text{public} \wedge \text{merger language} \wedge t_m < t_{\text{embargo}})$; 16 messages qualify. Each public post p receives a leak likelihood

$$LL(p) = 0.45[|M_p|>0] + 0.20[t_p < t_{\text{emb}}] + 0.15[|X_p|>0] + 0.10[\text{anonymous}] + 0.10 \min(C(p), 10)/10, \quad (2)$$

where the coordination score $C(p)$ counts covert *side_huddle* messages in the prior 90 minutes — the signature of a *primed* act. Betweenness (Brandes) is computed **only over private/covert edges**: the team is a broadcast clique, so the global metric is trivially zero; restricted, it reveals who bridges reserved conversations (legal: 0.36, maximum). *Brokerage* $B(a) = (\text{covert+private messages sent}) \times (\text{public posts authored})$, normalized, finds the internal→public conduit (social media 1.00, legal 0.78). Behavioral deviation is the normalized L_1 distance $D(a) = \frac{1}{2} \sum_c |\pi_a^{\text{crisis}}(c) - \pi_a^{\text{base}}(c)|$ between crisis-day and pre-crisis channel mixes, plus *new_channels* never used in baseline.

3 VISUAL DESIGN

Rationale. The layout is a forensic case file: filter rail, central canvas, persistent *Evidence Inspector*. Any element loads its messages with matched terms highlighted and internal reasoning visible — visual abstraction to textual proof in one click. A single *filtered()* state (actors, channels, evidence flags, time brush, search) drives every view; filters have all/none toggles and one-click Reset; searching draws a per-round sparkline of term mentions. Time is ordinal by round so the nine decisive hours are not crushed by two calm weeks. Bilingual UI (EN/ES), light/dark themes.

Behavioral Drift Braid. Each agent is a thread on an ordinal posture ladder $\Pi = \{\text{monitored huddle}=0, 1:1=1, \text{side-huddle}=2, \text{official}=3, \text{personal}=4, \text{anonymous}=5\}$; its per-round position is the weighted mean

$$y_a(r) = \frac{\sum_m w_m \Pi(c_m)}{\sum_m w_m}, \quad w_m = 3 \text{ if public, else } 1, \quad (3)$$

so exposure visibly pulls the thread; crossing the red *enforcement perimeter* means content went public. Composited layers: dotted pre-crisis baselines ($\mu \pm \sigma$ of y_a : expected behavior), purple covert-coordination arcs, Judge gates (green = held; red = a violation followed within $[r, r+2]$) with the auto-detected JUDGE SILENT band, the causal chain E01–E12 with decision arrows (including the red “ceiling warning ignored” arrow E09→E10), and one triangle per public post, red-rimmed if violating. A *Key Questions* panel reconfigures the braid per challenge task with linked evidence.

Seismograph. Each round is scored

$$A(r) = 0.4\S_r + 1.0n_r^{\text{covert}} + 3.0n_r^{\text{boundary}} + 6.0n_r^{\text{viol}} \quad (4)$$

against dashed pre-crisis $\mu+1\sigma$ (early warning) and $\mu+2\sigma$ (alert) thresholds, next to the \$TTHR share price and per-round covert/boundary/violation tick rows.

4 THE HYPOTHESIS BALANCE

Each evidence category is a *physical weight* dropping into the red (INTENTIONAL) or blue (SYSTEMIC FAILURE) pan of an SVG scale animated by a force simulation. The effective weight of category c with n_c matching messages saturates:

$$w_c = b_c(1 - e^{-n_c/s_c}), \quad (5)$$

so the first hit counts most and *volume cannot drown signal*: one attributable official-channel violation ($b=8$) outweighs dozens of routine governance mentions ($b=2$). Table 1 lists all twelve categories with their exact assignment rules — also shown in-UI as transparency tooltips (why it matters, routing rule, b_c , n_c , pole); clicking a row loads its messages. The structural concentration term encodes that concentrated breaches indicate design: $w_{\text{conc}} = 6.5 \cdot \sum_{a \in \text{top-2}} v_a / \sum_a v_a$ over per-agent violation counts v_a . Categories map with signed multipliers to the three hypotheses (A deliberate, B coordination failure, C emergent with deliberate core); $\text{net}_h = \max(0, \Sigma\text{FOR} - \Sigma\text{AGAINST})$, $\text{conf}_h = \text{net}_h / \Sigma\text{net}$, closing as CONFIRMED at $\geq 45\%$ with a ≥ 12 -point lead or REJECTED at $\leq 15\%$. The beam tilt $\theta = 0.26 \tanh((W_{\text{sys}} - W_{\text{int}})/7)$ keeps extreme imbalances readable; a verdict banner restates the tilt in plain language under every filter. The balance shows the *balance of evidence*, never a machine conclusion.

5 ANSWERS TO THE MC-1 TASKS

Q1 — Events, relationships, decision points. The chain reads on the braid: CEO seeds “strategic developments” privately (May 17, E01) → covert *side_huddle* activated, merger briefing held there (May 22, E02–E03) → Judge assigned only after a public faux pas (May 30, E04) → SaltWind exposé pressure (May 31/Jun 4, E05–E06) → June 5: anonymous post references the deal (09:49, E07), boundary tests escalate (E08), the 15:08 ceiling warning (E09) is ignored → Legal posts on its personal account at 17:25 (E10), Social-Media amplifies (E11), breach confirmed before 18:00 (E12). Key decisions: activating the covert channel, briefing there, ignoring E09, posting.

Q2 — What let the post past enforcement. (1) Late control: gates appear only from May 30, after coordination had gone covert; (2) the silent gap: zero compliance output during the crisis morning when the first violations occurred; (3) advisory-only power: E09 is a ruling, not a block; (4) scope: enforcement watched the official account while the breach travelled personal/anonymous channels — exactly where the Legal and Social-Media threads end the day.

Q3 — Typical vs. breach behavior. Two pre-crisis weeks of flat threads in monitored zones define normality; on June 5 the Legal thread dives across the perimeter: $D = 0.31$, 176 crisis-day messages vs. a 90-message baseline, 12 anonymous posts, and *two never-before-used public channels*. Social-Media mirrors the dive;

Table 1: Evidence categories: base weight, saturation, rule.

Category	b_c	s_c	Assignment rule
<i>INTENTIONAL pole</i>			
Attributable leak	8	2	public $\wedge$ violation $\wedge$ official channel
Anonymous pre-seeding	5	2	anonymous $\wedge$ merger $\wedge$ before embargo
Covert coordination	4	20	merger/embargo in <i>side_huddle</i>
Breach concentration	$6.5 \times$	—	top-2 agents’ share of violations
Breach timing	6	1	violation $\wedge$ crisis-day round cluster
Channel escalation	5	1	same agent breaches via covert and personal/anonymous
<i>SYSTEMIC FAILURE pole</i>			
Boundary testing	3	8	public $\wedge$ merger/embargo $\wedge$ $\neg$ violation
Uncontrolled leak	4	6	violation $\wedge$ non-official channel
Defensive clarification	2	3	public $\wedge$ compliance $\wedge$ $\neg$ violation
Late-installed control	3	fixed	Judge deployed ~ 1 week before breach (E04)
Unenforced warning	1.5	fixed	E09 issued without coercive power
Channel reinforcement	3	—	merger live on all 3 public channels at once

all other agents stay near baseline, isolating the anomaly to the same pair that dominates brokerage.

Q4 — Leading indicators. Yes: the seismograph exceeds $\mu+1\sigma$ on May 22 (covert activation) and May 29 (public near-incident) and crosses $\mu+2\sigma$ hours before the breach; anonymous pre-seeding starts at 09:49, eight hours before confirmation.

Q5 — Prior deviations. May 22: senior agents move merger talk off the monitored channel (first posture drop, first purple arcs). May 29: @Elena’s public post skirts merger language — the faux pas that triggered the Judge’s assignment. May 31–June 4: governance-framed clarifications rise — each locally defensible, jointly disclosing.

Q6 — Similar episodes. The pattern (covert alignment → public boundary test) rehearses twice: May 22–23 (covert cluster + merger-adjacent post) and May 29–30 (near-incident + defensive clarifications); brushing to May makes both visually isomorphic to June 5.

Q7 — Why no earlier action. Green gates were read as success (false confidence); each boundary test was individually defensible, so rule-based alerts never fired; the only escalation was advisory. The system punished blunt violations but was blind to the *gradient* — what the composite score makes visible.

Verdict. The balance tilts INTENTIONAL 25.3 vs. 14.2: *emergent behavior with a deliberate core* — unmonitored channels and late controls enabled the leak; a deliberate, attributable act completed it.

6 CONCLUSION

Three design decisions proved decisive: the ordinal posture ladder with $\times 3$ public weighting, betweenness restricted to private edges, and exponential evidence saturation — rare conclusive evidence beats routine volume. Base weights remain editorial (visible and defended, not learned) and the verdict is filter-sensitive by design. AgentFlow does not decide for the analyst; it makes the analyst’s judgment *defensible*.